PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $(B_1)_x = 0,\ (B_1)_y = \frac{\mu_0 I}{4\pi d}\left(-\hat{j}\right)$

$(B_2)_x = \frac{\mu_0 I}{8\pi d}\sin\theta\hat{i},\ (B_2)_y = \frac{\mu_0 I}{8\pi d}\cos\theta\left(-\hat{j}\right)$

$(B_3)_x = \frac{\mu_0 I}{8\pi d}\cos\theta\left(-\hat{i}\right),\ (B_3)_y = \frac{\mu_0 I}{8\pi d}\sin\theta\left(-\hat{j}\right)$

$B_{net} = \frac{\mu_0 I}{8\pi d}\left[\sin\theta - \cos\theta\right]\hat{i} - \frac{\mu_0 I}{8\pi d}\left(2 + \cos\theta + \sin\theta\right)\hat{j}$

$B_{net} = \frac{\mu_0 I}{8\pi d}\sqrt{6 + 4\sin\theta + 4\cos\theta}$

$f(\theta) = 6 + 4\sin\theta + 4\cos\theta$

$f'(\theta) = 4(\cos\theta - \sin\theta) = 0$

$\tan\theta = 1$ for $0^0 < \theta < \frac{\pi}{2}$

$\theta = \frac{\pi}{4}$

$\therefore$ **(a)**

2. Angular momentum of the body about any point on the body

$\vec{L}_P = I_P\vec{\omega} = \left(\frac{MR^2}{2} + \frac{MR^2}{4}\right)\omega = \frac{3MRv_0}{4}$

$\therefore$ **(d)**

3. At equilibrium $2N\cos 30^0 = mg$

$\Rightarrow \quad N = \frac{mg}{\sqrt{3}}$

$\frac{2N}{2} = \frac{Kl}{2}$

$\frac{K}{m} = \frac{2g}{\sqrt{3}l}$

Applying energy concepts $-mgx + \frac{1}{2}K\left(\frac{x}{\sqrt{3}}\right)^2 + \frac{1}{2}mv^2 = E$

Differentiating we get $\frac{dE}{dt} = 0$

$$-mgv+\frac{Kx}{3}v+mv\frac{dv}{dt}=0$$

So $a=\frac{-2g}{3\sqrt{3}l}x+mg$

$$T=2\pi\sqrt{\frac{3\sqrt{3}l}{2g}}$$

$\therefore$ **(b)**

4. $(F_b-F_A)x=\frac{Mg(y-x)}{2}$

where $F_b=m_bg=\rho_{al}Vg$ (force of gravity = force of Buoyancy)

and $F_A=\rho_w g\frac{V}{2}$ and $V=\frac{4}{3}\pi r^3$

$\therefore$ $\frac{y}{x}=1.5$

$\therefore$ **(a)**

5. $T=\left(\frac{T_2-T_1}{l}\right)x+T_1$ if $(T_2>T_1)$

$v=k\sqrt{T}$

or $\frac{dx}{dt}=k\sqrt{T}$

$$\int_0^l \frac{dx}{\sqrt{\left(\frac{T_2-T_1}{l}\right)x+T_1}}=\int_0^t kdt$$

$\therefore$ **(a)**

6. Given that $P=2e^{2V}$

$$\frac{nRT}{V}=2e^{2V}$$

$$T=\frac{2V}{nR}e^{2V}$$

By differentiating $\frac{dT}{dV}=\frac{2}{nR}\left[2Ve^{2V}+e^{2V}\right]$

$$C=C_V+\frac{PdV}{ndT}=C_V+\frac{R}{1+2V}$$

$\therefore$ **(a)**

7. At steady state of the capacitor

$$I = \frac{\varepsilon}{3R}$$

also $V_B - V_C = 2\varepsilon$ and $V_0 - V_B = 2\varepsilon$

$\therefore$ $E_1 = \frac{1}{2}C_1(2\varepsilon)^2 = 2C_1\varepsilon^2$

$$E_2 = \frac{1}{2}C_2\left(2\varepsilon - \frac{\varepsilon}{3}\right)^2 = \frac{25C_2\varepsilon^2}{18} \text{ now } E_1 = E_2$$

$$2C_1\varepsilon^2 = \frac{25C_2\varepsilon^2}{18}$$

$\therefore$ $\frac{C_1}{C_2} = \frac{25}{36}$

$\therefore$ **(a)**

8. By distribution of charge, all the capacitor of 1μF in between the above and lower branch is removed, then the equivalent capacitance of upper branch is

$$\frac{1}{C_{eq}} = \frac{1}{1} + \frac{1}{3} + \frac{1}{9} + ...$$

$$C_{eq} = \frac{1 - \frac{1}{3}}{1} = \frac{2}{3}$$

$\therefore$ $C_{net} = 2 \times \frac{2}{3} = \frac{4}{3}\mu F$

$\therefore$ **(b)**

SECTION – II
Multiple Correct Choice Type

9. $\frac{\Delta\theta}{\Delta x} = \frac{H}{AK}$

Heat flow is $\propto \frac{1}{AK}$

$\therefore$ **(a, c)**

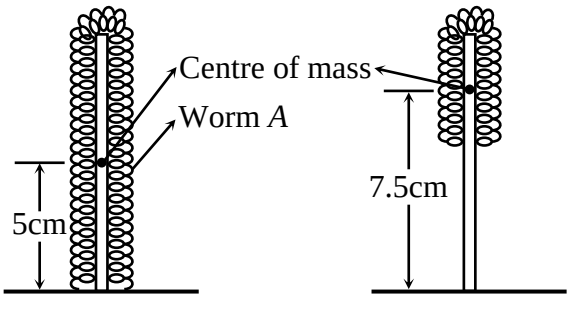
10. On worm *A* work done by gravity

$W_1 = -\Delta U_1$

For worm *B*, work done by worm

$W_2 = \Delta U_2$

$\therefore$ **(a, c)**



11. Visible range is from 3800 Å to 7800 Å and the wavelength 6000 Å corresponds to yellow colour so blue and violet colour can cause photoelectric emission, whatever be the intensity.

$\therefore$ **(b, c, d)**

12. For $\lambda = 1,\ dU + W = 0 \Rightarrow Q = 0$

For $\lambda = 0,\ dU = 0 \quad \Rightarrow$ T = constant

For $\lambda = -\frac{5}{2},\quad dU = \frac{5}{2}W$

$Q = dU + W = \frac{7}{2}W$

$\frac{Q}{dU} = \frac{7}{5} \quad \Rightarrow \quad Q = \frac{7}{5} \times n\frac{5}{2}RdT$

$Q = nC_p dT \quad \Rightarrow$ Pressure is constant

For $\lambda = \frac{5}{2}$,

$\Rightarrow \quad W = -\frac{2}{5}dU = \frac{-2}{5}(nCvdT)$

$W = -nRdT = \frac{P_2V_2 - P_1V_1}{-1}$

If PV^N = constant

$\Rightarrow \quad W = \int_{v_1}^{v_2} P\,dV = \frac{P_2V_2 - P_1V_1}{1-N}$

Solving $1 - N = -1$

$\Rightarrow \quad N = 2$

$\therefore$ **(a, c, d)**

SECTION–III
Comprehension Type
Passage-I

13. $2mv_0 = (2m + M)v' \Rightarrow v' = \frac{2v_0}{3}$

$\therefore$ **(d)**

14. $\frac{1}{2}(2m)v_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{1}{2}(3m)\left(\frac{2v_0}{3}\right)^2 + U$

$\therefore$ **(b)**

15. $2mv_0 = 2mx + m\left(\frac{v_0}{2}\right)\ ;\ \frac{3mv_0}{2} = 2mx\ ;\ x = \frac{3v_0}{4}$

$\frac{1}{2}(2m)v_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{q^2}{4\pi \in_0 x_1} + \frac{1}{2}(2m)\left(\frac{3v_0}{4}\right)^2 + \frac{1}{2}m\left(\frac{v_0}{2}\right)^2$

$mv_0^2 - \frac{11}{16}mv_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{q^2}{4\pi \in_0 x_1}$

$$\frac{5mv_0^2}{16}+\frac{q^2}{4\pi\in_0 r}=\frac{q^2}{4\pi\in_0 x_1}$$

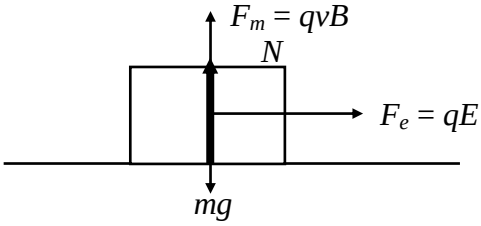
$$x_1=\frac{q^2}{4\pi\in_0\left(\frac{q^2}{4\pi\in_0 r}+\frac{5mv_0{}^2}{16}\right)}.$$

$\therefore$ **(a)**

Passage-II

Initially the charged block will experience gravitational and electric forces, but no magnetic force being at rest. Electric force will accelerate the block from rest.

At any instant when it moves



$F_m = qVB \left[\because \vec{V} \perp \vec{B}\right]$

$N + qvB = mg$

$N = mg - qvB = mg - q\left\{\frac{qE}{m}t\right\}B$ $\left[\because a = \frac{F_e}{m} \text{ and } v = at\right]$

$N = mg - \frac{q^2EBt}{m}$ … (i)

16. The block will lose contact when $N = 0$

From equation (i) $t = \frac{m^2 g}{q^2 EB}$

Putting values $t = 100$ sec

$\therefore$ **(c)**

17. Distance travelled upto $t = 100$ sec

$S = \frac{1}{2}\left\{\frac{qE}{m}\right\}t^2 = 5 \times 10^5$ m

$\therefore$ **(d)**

18. From equation (i) magnitude of time rate of change of N

$\left|\frac{dN}{dt}\right| = \frac{q^2EB}{m} = 10^{-2}$ N/sec

$\therefore$ **(a)**

SECTION–IV
Matrix–Match Type

1. **(A)** $\vec{v} = 5t\,\hat{i} + 4t\,\hat{j}$

$\vec{a} = \dfrac{d\vec{v}}{dt} = 5\hat{i} + 4\,\hat{j}$

So acceleration is constant

Also $\vec{v}\,(\text{at } t = 0) = 0$

So particle moves in a straight line.

(B) $\vec{v} = -5t\,\hat{i} + 5\,\hat{j}$

$\dfrac{d\vec{v}}{dt} = -5\hat{i}$ acceleration is constant and along $-x$ axis.

Now initial velocity is $5\,\hat{j}$

So particle moves in parabolic path.

(C) $y = 2x - 5x^2$

This is the equation of a parabola.

Hence path of the particle is parabolic. A particle which experience constant acceleration will follow parabolic path if projected in a direction other then the direction of acceleration.

(D) $x = 6\sin\dfrac{2\pi t}{5}$, $y = 6\cos\dfrac{2\pi t}{5}$

$x^2 + y^2 = 36$

Path is circular with constant speed.

$\therefore$ **(A) – (1, 3, 4), (B) – (2, 3, 4), (C) – (2, 3, 4), (D) – (4, 5)**

2. For He^+

$$E_n = \frac{-13.6\,Z^2}{n^2}\text{eV},\ E_1 = \frac{-13.6\times Z^2}{1^2} = -54.4\,\text{eV}$$

As reference has been shifted

$$E_n = \frac{-13.6\,Z^2}{n^2}\text{eV} + 54.4\,\text{eV}$$

For $n = 2,\ E_2 = \dfrac{-13.6\times Z^2}{2^2} + 54.4\,\text{eV} = 40.8\,\text{eV}$

Ionization energy from ground state is 54.4 eV

To excite an electron from $n = 1$ to $n = 2$

$\Delta E = -13.6\text{ eV} - (-54.4\text{ eV}) = 40.8\text{ eV}$

$$U_n = \frac{-27.2\,Z^2}{n^2} + 54.4\,\text{eV} = -54.4\,\text{eV}$$

$\therefore$ **(A) – (2, 3, 5), (B) – (1, 4), (C) – (2, 4, 5), (D) – (1, 3)**